



✦ Hi, I'm

Juan Carlos Ruiz García

Lead AI Engineer

PhD, Computer & Telecom Engineering ·
AI Specialization · Cum Laude

● Currently leading AI at JubilaME

"Turning AI into real products."

● CONTACT

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in linkedin.com/in/jcruiz-garcia
github.com/juanka1995
scholar.google.es/citations
📍 Spain · Madrid

● CORE TECH STACK

LANGUAGES & ML

Python Scikit-learn Pandas
NumPy Jupyter

CLOUD & BACKEND

GCP FastAPI Docker
Microservices SaaS

AUTOMATION & DEVOPS

n8n GitHub Actions GitLab CI/CD
Playwright

AI / DATA

Predictive Modeling Feature Selection
Time Series GenAI / LLMs NLP
Prompt Engineering

OTHER

MongoDB SQL Java Flask
Flet Linux / Bash



01

PROFILE

Lead AI Engineer with a PhD in Computer Science, specialized in bridging advanced analytical research with real-world tech infrastructure. Combines a strong algorithmic foundation with hands-on experience in cloud architecture (GCP), process automation, and ML productization. Proven track record leading R&D initiatives funded under the prestigious NEOTEC excellence program, managing multi-tenant SaaS infrastructure, and translating business requirements into scalable technical solutions. Backed by 4+ years of applied ML research in e-Health and HCI, plus active teaching experience at Master's level in Big Data and Business Intelligence.



02

PROFESSIONAL EXPERIENCE



JubilaME

Mar 2025 – Present

Full-time · Remote · Madrid, Spain

Lead AI Engineer

Aug 2025 – Present

- ▶ Took full ownership of the company's technological ecosystem and cloud infrastructure while continuing to drive advanced data-science initiatives.
- ▶ **Cloud Architecture:** managing and scaling GCP infrastructure for concurrent deployment of B2B platforms serving major institutional clients.
- ▶ **Backend & Microservices:** building robust microservices with FastAPI to create scalable, modular solutions across the company.
- ▶ **Business Automation:** leading end-to-end automation of business processes and IT operations using n8n and AI workflows (billing, reporting).
- ▶ **IT & Access Management:** improved internal security baseline (centralized password manager + 2FA). Administer Google Workspace.
- ▶ **Technical Leadership:** auditing external software providers, conducting code reviews, and translating business needs into technical execution.

Machine Learning & Data Scientist – Marketing

Mar 2025 – Aug 2025

- ▶ **Predictive Modeling:** designed ML models for customer micro-segmentation, optimizing marketing campaigns and personalized engagement.
- ▶ **Marketing Automation:** developed data-driven workflows to maximize ROI and improve targeting efficiency.
- ▶ **AI Integration:** integrated GenAI tools and LLM prompting into marketing strategies for content generation and engagement.
- ▶ **Prototyping & Analytics:** built Python prototypes and ran A/B testing for decision-making.

● LANGUAGES

Spanish Native
English Professional

● EDUCATION

PhD, Computer & Telecommunication Engineering

UAM · 2021 – 2025

Specialization in AI · Cum Laude

MSc in Research & Innovation in ICT

UAM · 2019 – 2021

Award for Excellence in MSc Thesis

BSc in Computer Engineering

Universidad de Granada · 2015 – 2019

● AWARDS

PhD · Cum Laude distinction

UAM · 2025

Excellence in Master's Thesis

UAM · 2021

Google Hash Code 2017

Global Rank #1437

Team "Los pollos hermanos" · Online qualification

● FUNDING LED

NEOTEC 2024 — CDTI

€325,000 · 85% funding ratio

Ministry of Science, Innovation & Universities (Spain). Code SNEO-20241008.

● CONFERENCE ORGANIZATION

ACM MobileHCI 2023

Athens, Greece

Local Organizer — Workshop on Advances of Mobile & Wearable Biometrics.

IEEE AVSS 2022

Madrid, Spain

Local Organizer — 18th IEEE Conference on Advanced Video & Signal-Based Surveillance. Sponsored by IEEE Computer Society PAMI TC & IEEE Signal Processing Society.

● RESEARCH GROUP

BiDA Lab

Universidad Autónoma de Madrid

Biometrics & Data Pattern Analytics — a leading Spanish research group in biometrics, HCI and applied ML.



UNIE Universidad

Sep 2023 – Sep 2025

Part-time · On-site · Madrid

- ▶ **Adjunct Professor — MSc Big Data & Business Intelligence (MaBDBI):** Advanced Statistics and Data Mining (regression, Bayesian learning, time series, recommender systems).
- ▶ **Adjunct Professor — BSc Computer Engineering:** user interfaces and UI prototyping.
- ▶ **Coordinator of External Internships** for MaBDBI and MSc Cybersecurity (MaCBS), Nov 2024 – Sep 2025.
- ▶ **Coordinator of the Master's Final Thesis** (MaCBS), Nov 2024 – Sep 2025.



Universidad Autónoma de Madrid

Sep 2019 – Dec 2024

Full-time · Hybrid

- **Research Data Scientist — AI for e-Health** Sep 2021 – Dec 2024
 - ▶ Designed health interaction tools and applied ML for Healthcare.
 - ▶ Developed quantitative metrics for assessing children's motor and cognitive development through tablet devices.
 - ▶ Built a Fall Detection System (FDS) for elderly people using smartwatch inertial sensors.
 - ▶ Authored multiple publications in IEEE Access, IEEE TETC and Pattern Recognition.
- **Research Assistant — Electronics & Communication Tech.** Mar 2020 – Sep 2021
 - ▶ Data analysis in e-Health and e-Learning research lines.
 - ▶ Joined BiDA Lab (Biometrics and Data Pattern Analytics).
- **Research Assistant — Computer Engineering** Sep 2019 – Mar 2020
 - ▶ Data Scientist in Learning Analytics; mining of MOOC data on the edX platform.



Granada Dynamics

Mar 2019 – Sep 2019

Internship · On-site · Granada

- **Python Junior Developer**
 - ▶ Testing tasks and maintenance of NoSQL / MongoDB databases.
 - ▶ Development of conversational agents and NLP pipelines.
 - ▶ Web services and REST API development.

● AT A GLANCE

€325K R&D funding led (NEOTEC)

10+ peer-reviewed publications

4+ years applied ML research

2 int'l conferences organized

2 years university teaching

● DOMAINS

FinTech InsurTech e-Health
e-Learning HCI Wearables
Optimization SaaS B2B

● BEYOND WORK

● Football ● Cycling ■ Padel
▲ Hiking ◆ Gaming
▲ Board games ◆ Friends

● INTERNATIONAL PUBLICATIONS

- 1 D. Jimenez Oviedo, R. Vera-Rodriguez, R. Tolosana, J.C. Ruiz-Garcia, J. Herreros-Rodriguez. Longitudinal Digital Phenotyping for Early Cognitive-Motor Screening. in *Proc. IEEE Conference on Artificial Intelligence*, Granada, Spain, May 2026.
- 2 J.C. Ruiz-Garcia, R. Tolosana, R. Vera-Rodriguez, A. Morales, J. Fierrez, J. Ortega-Garcia and J. Herreros-Rodriguez. Longitudinal Analysis and Quantitative Assessment of Child Development through Mobile Interaction. *IEEE Access*, 2024.
- 3 J.C. Ruiz-Garcia, R. Tolosana, R. Vera-Rodriguez, J. Fierrez and J. Herreros-Rodriguez. ChildCI Framework: Analysis of Motor and Cognitive Development in Children-Computer Interaction for Age Detection. *Cognitive Systems Research*, 2024.
- 4 J.C. Ruiz-Garcia, C. Hojas, R. Tolosana, R. Vera-Rodriguez, A. Morales, J. Fierrez, J. Ortega-Garcia and J. Herreros-Rodriguez. Children Age Group Detection based on Human-Computer Interaction and Time Series Analysis. *International Journal on Document Analysis and Recognition*, 2024.
- 5 J.C. Ruiz-Garcia, R. Tolosana, R. Vera-Rodriguez, A. Morales, J. Fierrez, J. Ortega-Garcia and J. Herreros-Rodriguez. ChildCldb_v2: A Longitudinal Database for Children-Computer Interaction on Mobile Devices. in *Proc. ACM International Conference on Mobile Human-Computer Interaction Workshop, MobileHCIw*, Athens, Greece, September 2023.

03 KEY PROJECTS

NEOTEC Excellence Program 2024

Mar 2025 – Present

JubilaME × CDTI · AI-Driven Financial & Pensional Planning · €325K

Code: SNE0-20241008

Link: cdti.es/ayudas/ayudas-neotec-2024

- ▶ High-impact collaboration with CDTI under the prestigious NEOTEC program, focused on the digital transformation of the Silver Economy.
- ▶ Designed predictive models for micro-segmentation and financial behavior analysis powering a hybrid B2B / B2B2C SaaS platform.
- ▶ Deployed a scalable, multi-tenant architecture on GCP with robust encryption and GDPR-compliant identity management.
- ▶ Led technical execution for the "Bono Doctor" excellence certification, optimizing the public funding ratio to 85%.

GCP FastAPI Predictive Modeling SaaS FinTech

Gemini AI Betting Bot

Feb 2026

End-to-End Automation & Intelligent Analysis

Repo: github.com/juanka1995/apuestas-bot

- ▶ Integrated Google Gemini "Deep Research" with Python / Playwright to generate portfolios automatically.
- ▶ Orchestrated 4+ automated workflows via GitHub Actions (cron-based); containerized with Docker.
- ▶ Built a bespoke Chrome extension (Manifest V3) for automated cookie rotation and secure credential management.

Python Playwright GitHub Actions Docker LLMs

ChildCldb — Child-Computer Interaction Database

Dec 2019 – Dec 2024

Universidad Autónoma de Madrid · BiDA Lab

Repo: github.com/BiDALab/ChildCldbLong

- ▶ The largest publicly available CCI dataset to date — 615+ children aged 18 months to 8 years, 2.1K+ sessions and 12.6K+ tests across 4 academic years (2019/20 – 2022/23).
- ▶ Designed acquisition protocols, mobile data collection app, and longitudinal curation pipeline.
- ▶ Foundation for 5+ peer-reviewed publications in IEEE journals.

Mobile Longitudinal HCI e-Health IEEE

CareFall — Automatic Fall Detection

May 2022 – Dec 2023

BiDA Lab × Cartronic Group · University-Enterprise collaboration

- ▶ Fall Detection System processing 3-axis accelerometer and gyroscope signals from commercial smartwatches.
- ▶ Implemented and benchmarked threshold-based (SMV, Fall Index) vs ML-based classification approaches.
- ▶ ML approach significantly outperformed traditional methods on public datasets in accuracy, sensitivity and specificity.

Wearables Signal Processing ML e-Health Smartwatch

- 6 J.C. Ruiz-Garcia, R. Tolosana, R. Vera-Rodriguez, C. Moro. CareFall: Automatic Fall Detection through Wearable Devices and AI Methods. in *Proc. ACM International Conference on Mobile Human-Computer Interaction Workshop, MobileHCIw*, Athens, Greece, September 2023.
- 7 R. Tolosana, J.C. Ruiz-Garcia, R. Vera-Rodriguez, J. Herreros-Rodriguez, S. Romero-Tapiador, A. Morales and J. Fierrez. Child-Computer Interaction with Mobile Devices: Recent Works, New Dataset, and Age Detection. *IEEE Transactions on Emerging Topics in Computing*, pp. 1-13, 2022.
- 8 R. Tolosana, R. Vera-Rodriguez, C. Gonzalez-Garcia, J. Fierrez, A. Morales, J. Ortega-Garcia, J.C. Ruiz-Garcia, et al. SVC-onGoing: Signature Verification Competition. *Pattern Recognition*, Vol. 127, 108609, 2022.
- 9 R. Cobos, J.C. Ruiz-Garcia. Improving learner engagement in MOOCs using a learning intervention system: A research study in engineering education. *Computer Applications in Engineering Education*. 2021; 29: 733–749.
- 10 R. Tolosana, R. Vera-Rodriguez, C. Gonzalez-Garcia, J. Fierrez, S. Rengifo, A. Morales, J. Ortega-Garcia, J.C. Ruiz-Garcia et al. ICDAR 2021 Competition on On-Line Signature Verification. in *Proc. International Conference on Document Analysis and Recognition, ICDAR*, Lausanne, Switzerland, September 2021.

FUTCheap — AI-Powered FIFA Squad Builder

Apr 2017 – Jan 2023

Metaheuristic optimization for Ultimate Team squads

Link: futchep.com

- ▶ Application using metaheuristic algorithms to build optimal FIFA Ultimate Team squads under customizable constraints (price, chemistry, league, formation).
- ▶ Co-developed the foundational early phases alongside the lead developer.
- ▶ Per-position weighting, weak-foot/skills configuration, budget optimisation to maximize team quality at minimum coin cost.

AI

Optimization

Metaheuristics

Java

Python

Genetic Algorithm for Feature Selection

Jan 2021 – Apr 2022

BiDA Lab · github.com/BiDALab/GeneticAlgorithm

- ▶ Python framework for evolutionary feature selection (tournament selection, crossover, mutation, intelligent early-stopping).
- ▶ Modular and parallelizable, compatible with any Scikit-learn estimator.
- ▶ Core component of IEEE Transactions on Emerging Topics in Computing publication.

Python

Scikit-learn

Evolutionary Algorithms

Feature Selection

IEEE